llmXive Automated Review of SANA-WM: Efficient Minute-Scale World Modeling with Hybrid Linear Diffusion Transformer

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong factual claims regarding performance metrics and hardware specifications that require verification or clarification to ensure strict accuracy.

First, the Abstract and Introduction claim that a distilled variant can deploy on a "single RTX 5090" to generate a 60s clip in 34s. As of the current date, the NVIDIA RTX 5090 is not a publicly available or benchmarked product. Citing specific performance metrics on unreleased hardware renders this claim factually unverifiable and potentially misleading. The authors should either provide projected estimates clearly labeled as such, or rephrase the claim to reference available hardware (e.g., RTX 4090) with corresponding data.

Second, the claim of "$36\times$ higher throughput" in the Abstract and Introduction is not immediately supported by the visible data in Table 1. The table lists SANA-WM (refined) at 22.0 videos/hour and LingBot-World at 0.6 videos/hour. The ratio $22.0 / 0.6 \approx 36.6$, which rounds to 37x, or if comparing the non-refined SANA-WM (24.1) to LingBot (0.6), the ratio is $\approx 40x$. The specific "36x" figure appears to be a precise calculation that does not align perfectly with the rounded numbers in the main table, or it refers to a specific baseline configuration not explicitly highlighted in the text. The authors should clarify the exact baseline and calculation used to support this specific multiplier.

Finally, the abstract cites "~213K" clips, while Table 1 (tables/train-data.tex) provides an exact count of 212,975. While this is a minor discrepancy, scientific writing should maintain consistency between rounded claims and exact data tables. Using the exact number or explicitly stating the rounding method would improve precision.

The core scientific claims regarding the architecture (Hybrid Linear Attention, Dual-Branch Camera Control) and the general trend of improved efficiency and accuracy appear supported by the provided tables and ablation studies, but the specific hardware and throughput numbers need the adjustments noted above to be factually rigorous.

Action items.

- **[writing]** The claim that the distilled variant generates a 60s 720p clip in 34s on a single RTX 5090 (Abstract, Sec 1) relies on hardware (RTX 5090) that is not yet publicly released or benchmarked. This specific performance metric is currently unverifiable and should be qualified as 'projected' or replaced with data from an available GPU (e.g., RTX 4090) to ensure factual accuracy.
- **[writing]** The claim of '36x higher throughput' compared to baselines (Abstract, Sec 1) lacks a clearly defined baseline in the text. While Table 1 shows SANA-WM at 24.1 videos/hour, the text does not explicitly state which baseline model and configuration (e.g., LingBot-World at 0.6 videos/hour) yields exactly 36x. The calculation 24.1/0.6 40x suggests the '36x' figure may be rounded or derived from a different metric not explicitly shown, requiring

clarification to support the specific multiplier.

- **[writing]** The abstract states the model uses '~213K public video clips', but Table 1 (tables/train-data.tex) lists a total of 212,975 clips. While the difference is negligible, the text should either use the exact number or explicitly state the rounding convention to maintain precision in factual claims.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 effectively communicates efficiency gains but has issues with visual scaling in (a) that may mislead readers about the magnitude of improvements, and the 'OOM' notation lacks clear definition. Additionally, the continuous line to the 'OOM' point in (b) is misleading as it implies a valid measurement where none exists.

Figure 2

The figure provides a clear visual overview of the SANA-WM architecture and its components. However, the caption's description of token flow is slightly imprecise compared to the diagram, and there is a minor inconsistency between the main pipeline diagram and the detailed block view regarding where specific inputs (Text, Pose) are processed.

Figure 3

Figure 3 is critically flawed as it lacks a caption entirely, rendering the data uninterpretable. Furthermore, the plot displays 'NaN' values and undefined legend terms ('No Scale', 'DS'), suggesting severe data or reporting errors.

Figure 4

The figure displays a clear qualitative comparison of video generation methods over time, but the caption contains a critical grammatical error where the definition of the green borders is missing, rendering the annotation instructions incomplete.

Figure 5

The figure contains a critical grammatical error in the caption where the definition for the green borders is missing. Additionally, the generated videos in the bottom row fail to match the specific details (green tractor, fence) described in the provided prompt text.

Figure 6

Figure 6 is a clear and effective teaser that visually demonstrates the model's capabilities (minute-scale generation, action control) and efficiency claims (64-GPU training, 1-H100 GPU inference) as described in the caption. The layout is uncluttered, and the visual elements (video frames, timeline, icons) are legible and well-integrated to support the paper's main selling points.

Figure 7

Figure 7 is a clear and well-structured pipeline diagram that effectively visualizes the data construction process described in the caption. All stages, inputs, and outputs are clearly labeled with appropriate icons and text, and the flow is easy to follow.

Figure 8

Figure 8 effectively demonstrates the visual improvements of the proposed refiner over the base SANA-WM model across long-horizon rollouts. The layout is clear, with time progression indicated at the top and method variants labeled on the left, while red boxes successfully highlight specific regions of enhanced fidelity and consistency.

Figure 9

Figure 9 effectively displays representative initial frames across four scene categories (Game, Indoor, City, Nature) as described in the caption. The layout is clear, the categories are well-labeled, and the images are legible, successfully illustrating the diversity of the benchmark dataset.

Figure 10

Figure 10 effectively visualizes the benchmark trajectory templates with clear ground projections and corresponding height profiles. The caption accurately describes the 'Simple' and 'Hard' splits and the visual encoding of the data, and the figure is well-organized with legible labels and a clear legend.

Figure 11

The figure provides a visual demonstration of 3D reconstruction but suffers from a grammatical error in the caption and lacks necessary scale references to substantiate geometric claims.
Action items.

- **[science]** Figure 1a: The caption states ‘bars are scaled for readability,’ but the chart displays raw latency values (e.g., 21.7 min) alongside a ‘38x overall’ speedup claim. If the bars are scaled, the visual length does not represent the true magnitude of the difference, making the ‘38x’ claim visually misleading without a clear explanation of the scaling factor applied.
- **[writing]** Figure 1a: The text ‘OOM’ (Out of Memory) is used to indicate failure for the ‘w/o sink’ configuration, but there is no legend or explicit definition in the caption explaining that ‘OOM’ means the experiment failed to complete, which may be unclear to some readers.
- **[science]** Figure 1b: The ‘OOM’ point for the ‘Softmax only’ method is plotted at 60s video duration, but the line connecting the previous point to ‘OOM’ suggests a continuous trend. This is misleading because the method did not actually run at 60s; the plot should indicate a break or discontinuity to avoid implying a valid data point exists.
- **[writing]** Figure 2: The caption states ‘Text, video, and pose tokens pass through alternating GDN and softmax attention blocks,’ but the diagram shows ‘Text Tokens’ and ‘Camera Pose’ entering the GDN Blocks while ‘Ref-Image Tokens’ and ‘Latent Tokens’ enter the Softmax Block; the caption should clarify the specific input routing for each block type.
- **[writing]** Figure 2: The ‘GDN / SOFTMAX BLOCK’ detail view shows ‘Text Tokens’ entering ‘Cross Attention’ and ‘Camera Pose’ entering ‘Plucker Mixing’, but the main pipeline diagram shows ‘Text Tokens’ and ‘Camera Pose’ entering the ‘GDN Block’ directly; the diagram and detail view are inconsistent regarding where these inputs are processed.
- **[fatal]** Figure 3: The figure has no caption (explicitly labeled ‘(no caption)’), making it impossible to interpret the context, methods, or significance of the data shown.
- **[science]** Figure 3: The legend entry ‘No Scale’ is undefined and likely a placeholder; it does not describe a valid experimental condition or baseline.
- **[science]** Figure 3: The plot contains explicit ‘NaN’ annotations (‘NaN @ 1st step’, ‘NaN @ 16th step’), indicating data corruption or model failure, yet no explanation or analysis is provided.
- **[writing]** Figure 3: The legend entry ‘Ours (1/√DS)’ contains a likely typo or undefined variable ‘S’ which is not explained in the absence of a caption.
- **[science]** Figure 4: The caption states ‘Green borders mark , with transparent action overlays...’ but the phrase ‘mark’ is followed by a comma and nothing else, leaving the meaning of the green borders undefined.
- **[writing]** Figure 4: The caption contains a grammatical error and incomplete sentence structure (‘mark , with’) that obscures the intended comparison or metric being highlighted by the green borders.
- **[fatal]** Figure 5: The caption states ‘Green borders denote , with transparent action overlays...’ but the phrase following ‘denote’ is missing, leaving the meaning of the green borders undefined.

- [**science**] Figure 5: The prompt text on the left describes a 'green tractor' and 'fenced pasture', but the generated video frames (especially in the bottom row) show a red tractor and no fence, indicating a failure to follow the prompt which is not explained in the caption.
- [**writing**] Figure 11: The caption contains a grammatical error and missing noun in the phrase 'generated by .', leaving the subject of the generation undefined.
- [**science**] Figure 11: The figure displays 3D point cloud reconstructions but lacks any scale bars, coordinate axes, or metric references, making it impossible to verify the 'metric-scale' or geometric accuracy claims implied by the text.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript is dense with specialized acronyms and domain-specific terminology that significantly raises the barrier to entry for non-specialist readers. While the technical depth is appropriate for the field, the lack of definitions for standard acronyms violates the principle of accessibility.

Specifically, the Abstract introduces "6-DoF," "GDN," and "NVFP4" without definition. "6-DoF" should be spelled out as "six degrees of freedom" at first mention. "GDN" (Gated DeltaNet) is a core architectural component and must be defined immediately. "NVFP4" is a proprietary quantization format that requires a brief explanation or expansion (e.g., "NVIDIA FP4 quantization") to be understood by a broader audience.

In Section 3, "DiT" is used repeatedly for "Diffusion Transformer" without prior expansion. Similarly, "VAE" (Variational Autoencoder) and "UCPE" (Unified Camera Positional Encoding) appear without definition. The term "Pl\"ucker" (referring to Plücker coordinates) is used in Section 3.2; while standard in computer vision, it is obscure to generalists and should be briefly contextualized (e.g., "Plücker ray coordinates").

Section 4 introduces "3DGS" (3D Gaussian Splatting) and "FCGS" without expansion. Section 5 relies heavily on "AR" (autoregressive) and "FVD" (Fréchet Video Distance) without defining them first. The Appendix uses "LoRA" (Low-Rank Adaptation) without definition.

To improve readability, every acronym should be defined at its first occurrence in the text (Abstract, Introduction, or Method). Additionally, terms like "Plücker," "RoPE" (Rotary Positional Embedding, used in Eq. 3.2), and "FSDP2" should be either expanded or briefly explained in a footnote or parenthetical to ensure the paper is accessible to researchers outside the immediate sub-field of efficient video generation.

Action items.

- [**writing**] The manuscript is dense with specialized acronyms and domain-specific terminology that significantly raises the barrier to entry for non-specialist readers. While the technical depth is appropriate for the field, the lack of definitions for standard acronyms violates the principle of accessibility. Specifically, the Abstract introduces “6-DoF,” “GDN,” and “NVFP4” without definition. “6-DoF” should be spelled out as “six degrees of freedom” at first mention. “GDN” (Gated DeltaNet) is a core archi

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent architecture for minute-scale world modeling, but several causal links between proposed mechanisms and reported outcomes require clarification to ensure logical consistency.

First, the claim that the **Second-Stage Refiner** improves **camera-control accuracy** (Table 1 and Appendix Table A1 show reduced RotErr/TransErr for the refined model) is not logically supported by the described mechanism. The refiner is trained via truncated-σ flow matching to map noisy latents to high-fidelity targets, conditioned on text and camera poses. While this improves visual quality (VBench scores), the text does not explain *how* a visual refinement process reduces geometric pose error. If the Stage-1 model already generates a video with a specific camera trajectory, a visual refiner should ideally preserve that trajectory while sharpening details. A reduction in pose error suggests the Stage-1 model’s trajectory estimation (via Pi3X) was biased by visual artifacts, or the refiner implicitly corrects geometric drift. The authors must explicitly state whether the pose improvement is a secondary effect of reduced visual noise (making pose estimation more accurate) or if the refiner actively optimizes for geometric consistency. Without this distinction, the causal claim that the refiner “improves action following” is ambiguous.

Second, the **Dual-Branch Camera Control** mechanism (Sec 3.3) asserts that the Plücker mixing branch restores fine motion “inside each VAE stride.” The text describes this branch as a “zero-initialized per-block projection” added “immediately after each self-attention output.” Logically, adding a residual vector to the transformer output at the latent frame rate cannot recover high-frequency motion that was discarded by the VAE’s temporal downsampling (which aggregates 8 raw frames into 1 latent). If the VAE has already compressed the temporal resolution, the transformer output operates at the lower resolution. The text implies the Plücker branch operates at the “raw-frame” rate, but the implementation description places it as a block-wise addition to the latent stream. The authors need to clarify the signal flow: does the Plücker branch modulate the latent tokens to *predict* high-frequency details, or is there a separate upsampling step where this information is injected? The current description creates a logical gap between the “raw-frame” claim and the “block-wise addition” implementation.

Third, the **GDN Key Scaling** ablation (Fig 5) claims that unscaled keys lead to an expansive transition matrix $\mathbf{M}_t$, causing instability. The proposed fix scales keys by

$1/\sqrt{D \cdot S}$. The logic relies on the trace of the key outer product being bounded. However, the text states $\operatorname{tr}(\mathbf{A}_t) \le 1$ implies $\|\mathbf{M}_t\|_2 \le \gamma_t \le 1$. This assumes the spectral norm is bounded by the trace, which is true for positive semidefinite matrices, but the derivation assumes the gates β and normalized keys behave in a way that strictly satisfies this bound under all training dynamics. The paper asserts this scaling "ensures stable convergence" without providing the mathematical bound or counter-example for why $1/\sqrt{D}$ (standard token-wise scaling) fails specifically for the frame-wise aggregation of S tokens. A brief derivation or citation of the spectral bound for this specific frame-wise aggregation would strengthen the logical consistency of this design choice.

Action items.

- [**science**] The claim that the refiner improves 'camera-control accuracy' (Tab. 1) lacks a mechanistic explanation. The refiner is trained on visual fidelity, yet pose error (RotErr) drops. Clarify if this is a side effect of better visual consistency aiding pose estimation or if the refiner explicitly optimizes geometry.
- [**science**] The 'Dual-Branch Camera Control' section claims the Plücker branch restores motion 'inside each VAE stride' (Sec 3.3), yet it is added as a residual after self-attention. Explain how a block-wise addition to latent tokens recovers high-frequency motion discarded by VAE temporal downsampling.
- [**science**] The GDN key scaling ablation (Fig 5) asserts $1/\sqrt{D \cdot S}$ prevents matrix expansion but lacks a derivation. Justify why this specific scaling ensures the spectral radius of the transition matrix remains ≤ 1 for all gate values and key distributions.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding efficiency and performance that appear to extrapolate beyond the provided data or rely on unverified hardware specifications.

First, the Abstract claims the model achieves "36x higher throughput" for scalable world modeling. While the paper provides throughput numbers for SANA-WM (22.0 videos/hr) and lists baselines, the comparison is not apples-to-apples. The baselines listed (e.g., LingBot-World, HY-WorldPlay) are either 480p or require 8 GPUs. The claim of a 36x improvement implies a direct comparison to a 720p, single-GPU baseline, which is not explicitly quantified in the tables. Without a clear baseline definition for this specific multiplier, the claim risks over-claiming the magnitude of the efficiency gain.

Second, the Abstract states the distilled variant can deploy on a "single RTX 5090" to generate a clip in 34s. The RTX 5090 is a future, unreleased product at the time of writing. Presenting specific latency numbers (34s) for hardware that does not yet exist as a definitive result is an overreach. This should be framed as a projection based on theoretical compute or a benchmark on a comparable current-generation card (e.g., RTX 4090) with a scaling factor, rather than a measured fact.

Third, the claim of "visual quality comparable to... LingBot-World" is nuanced. Table 1 shows LingBot-World achieving a VBench Overall score of 81.82 (480p) while SANA-WM + Refiner achieves 80.62 (720p). While the resolution difference is significant, the text asserts comparability without fully addressing that the 480p baseline actually scores higher on the aggregate metric used. The authors should clarify if "comparable" refers to perceptual fidelity despite the metric gap or if the claim needs to be tempered to reflect that the 720p model is slightly behind the 480p baseline in specific VBench dimensions.

Finally, the claim of "precise 6-DoF trajectory adherence" is supported by RotErr/TransErr metrics, but the paper does not provide a statistical significance test or error bars on these metrics to justify the absolute term "precise" against the variance inherent in pose estimation (Pi3X) and generation. While the ablation studies show improvement, the absolute precision claim is strong given the reliance on estimated ground truth for evaluation.

Action items.

- [**writing**] The paper makes several strong claims regarding efficiency and performance that appear to extrapolate beyond the provided data or rely on unverified hardware specifications. First, the Abstract claims the model achieves "36x higher throughput" for scalable world modeling. While the paper provides throughput numbers for SANA-WM (22.0 videos/hr) and lists baselines, the comparison is not apples-to-apples. The baselines listed (e.g., LingBot-World, HY-WorldPlay) are either 480p or require 8 GPUs. T

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript addresses safety and ethics primarily in the "Broader Impact" section (Sec. 7.1) and the "Existing Assets and Tool Terms" table (Tab. 1 in Appendix). While the authors correctly identify the dual-use risks associated with high-fidelity video generation (e.g., deepfakes, simulation over-interpretation), the review identifies three specific areas where the safety and ethical disclosure requires strengthening before the paper can be considered fully compliant with responsible AI release standards.

First, there is a potential licensing conflict regarding the "open-source" claim. The paper states the model is open-source and accessible (Abstract, Sec. 1), yet the data pipeline relies heavily on assets with restrictive terms, such as SpatialVID-HQ (CC-BY-NC-SA 4.0), DL3DV (Custom terms), and Pi3X (CC BY-NC 4.0 for weights). The "Existing Assets" table lists these restrictions but does not explicitly clarify if the final released model weights or the generated benchmark data inherit these Non-Commercial (NC) constraints. If the model is trained on NC data, releasing it for commercial use could violate the source licenses. The authors must explicitly state the license of the final artifact and whether it is restricted to non-commercial research use only.

Second, the mitigation strategies for the identified risks are generic. The "Broader Impact" section mentions the need for "watermarks or other provenance signals" but does not confirm if the authors have implemented them. Given the model's capability to generate minute-long, 720p videos with precise camera control, the potential for creating convincing disinformation is significant. The review requires a definitive statement on whether the released model includes embedded watermarking (e.g., SynthID, as mentioned for the benchmark images) or if a specific usage policy (e.g., prohibiting political manipulation) will be enforced.

Third, the data privacy aspect of the "Robust Annotation Pipeline" (Sec. 4) is under-specified. The pipeline processes "public video clips" from sources like SpatialVID and MiraData. While the paper details filtering for aesthetic quality and motion, it does not mention any steps taken to detect and remove Personally Identifiable Information (PII), such as faces, license plates, or private locations, which are common in internet video datasets. Training on such data without consent or scrubbing poses privacy risks. The authors should clarify if a PII scrubbing or face-blurring step was integrated into the data preparation process.

Addressing these points will ensure the paper provides a complete picture of the ethical and legal implications of releasing such a powerful world model.

Action items.

- [**writing**] The 'Existing Assets and Tool Terms' table (Tab. 1 in Appendix) lists several datasets (e.g., SpatialVID-HQ, DL3DV) and models (e.g., Pi3X, LTX-2) with non-commercial (NC) or gated access terms. The paper claims the model is 'open-source' and 'accessible' but does not explicitly state whether the final trained weights or the generated benchmark data are subject to these same NC restrictions. Clarify the licensing status of the released artifacts to prevent legal ambiguity for downstream users.
- [**writing**] The 'Broader Impact' section (Sec. 7.1) acknowledges risks of deepfakes and simulation misuse but lacks a concrete mitigation strategy for the specific 'minute-scale' capability. Given the high fidelity and long duration, the risk of generating convincing disinformation is elevated. Explicitly state if the model weights will include watermarking (e.g., SynthID, Stable Signature) or if a usage policy prohibiting political/defamatory generation will be enforced upon release.
- [**writing**] The data pipeline relies on 'public video sources' (Sec. 4) and automated filtering. While the paper mentions filtering for 'visual quality,' it does not address the potential for the training data to contain sensitive personal information (PII), faces, or copyrighted material that was not explicitly consented to for AI training. Add a statement confirming whether a PII scrubbing or face-blurring step was included in the annotation pipeline to mitigate privacy risks.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling architecture for minute-scale world modeling, supported by extensive ablation studies on architectural components (GDN vs. Softmax, camera conditioning branches) and training stability (key scaling). The evidence for the efficacy of the hybrid attention mechanism and the specific key-scaling strategy (Eq. 8) is robust, with clear quantitative proof of stability improvements (Fig. 5, right).

However, the strength of the central claims regarding efficiency and benchmark performance requires further scrutiny. The claim of "$36\times$ higher throughput" in the Abstract is not grounded in a clearly defined baseline within the text or tables. While Table 1 lists various baselines, none are explicitly cited as the reference for this specific multiplier. Without a defined baseline (model, resolution, hardware), this claim is difficult to verify and risks being an overstatement.

Furthermore, the evaluation benchmark relies on 80 initial images generated by a proprietary model ("Nano Banana Pro"). The results presented in Table 1 and the Appendix are single-run averages. Given the stochastic nature of diffusion models and the sensitivity of long-horizon generation to initial conditions, the lack of variance reporting (e.g., standard deviation across multiple seeds or different initial image sets) makes it difficult to assess the statistical significance of the reported improvements over baselines like LingBot-World or HY-WorldPlay.

Finally, the ablation on the second-stage refiner (Appendix Table 1) reveals a notable trade-off: while the custom refiner improves visual quality (VBench), the "Original LTX-2.3" refiner actually achieves better Pose Accuracy (RotErr 8.65 vs 4.50 on Simple split). The paper frames the refiner primarily as a quality enhancer but does not adequately discuss this potential conflict between visual fidelity and geometric precision, which is critical for a "world model" intended for simulation. The authors should clarify the optimization objective of the refiner and whether the reported pose accuracy gains are a direct result of the refiner or the stage-1 model alone.

Action items.

- **[science]** The claim of '36x higher throughput' (Abstract) lacks a defined baseline. Specify the exact baseline model, resolution, and hardware configuration used for this comparison to allow independent verification of the efficiency claim.
- **[science]** The benchmark relies on 80 synthetic images generated by 'Nano Banana Pro' (Sec 5.2). The paper does not report variance across different random seeds or initial image sets. Re-run the benchmark with multiple seeds to ensure results are not driven by specific favorable initial conditions.
- **[science]** The 'Refiner' ablation (App. Tab. 1) shows a significant drop in Pose Accuracy (RotErr) when using the original LTX-2.3 refiner. The main text claims the refiner improves 'visual fidelity' but does not explicitly discuss this trade-off with action-following accuracy.

Clarify if the refiner is optimized for visual quality at the cost of geometric precision.

- [**science**] The training data includes 14,881 synthetic clips from DL3DV 3DGS rendering (Tab. 4). The paper does not quantify the domain gap between these synthetic clips and the real-world data. Provide an ablation or analysis showing the impact of removing synthetic data on the final model's generalization to real-world trajectories.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical rigor of the experimental evaluation requires strengthening to support the paper's strong claims regarding performance and stability.

First, the ablation study on GDN key scaling (Sec. 5.4, Fig. 5) presents a binary outcome: specific scaling factors lead to "immediate gradient instability" and "NaN events," while the proposed $1/\sqrt{DS}$ scaling ensures "stable convergence." However, the analysis lacks statistical aggregation. Deep learning training is inherently stochastic; a single run showing NaNs could be an outlier, or a single stable run could be lucky. The authors should report these stability results as an average over at least 3 independent random seeds, including the standard deviation of the final loss or the failure rate (e.g., "3/3 seeds converged vs. 0/3 seeds for baseline"). Without this, the claim of "immediate" instability is anecdotal rather than statistical.

Second, the main benchmark results (Sec. 5.2, Tab. 1) are based on a protocol that generates "one generated video per benchmark scene." While the benchmark contains 80 scenes, the metrics (RotErr, TransErr, VBench scores) are aggregated without reporting the variance across these scenes or across multiple model seeds. For instance, the improvement in RotErr from 10.02° to 8.34° (Hard split) is presented as a definitive win. To validate this, the authors should report the mean and standard deviation of these metrics across the 80 scenes. Furthermore, if the model generation involves stochastic sampling (which diffusion models do), the results should ideally be averaged over multiple seeds per scene to ensure the reported gains are not due to favorable random seeds.

Finally, the efficiency claims (Abstract, Tab. 1) cite specific throughput numbers (e.g., "24.1 videos/hour") and memory usage. These are presented as point estimates. Given that GPU performance can vary due to thermal throttling, background processes, or kernel launch overhead, reporting the standard deviation of latency and throughput over multiple inference runs (e.g., 5-10 runs) would provide a more robust statistical basis for the "36x" efficiency claim.

In summary, while the experimental design is sound, the lack of variance reporting (standard deviations, confidence intervals, or multi-seed aggregation) makes it difficult to assess the statistical significance of the reported improvements and the reliability of the stability claims.

Action items.

- **[science]** The ablation study in Fig. 5 (loss-vs-scale.pdf) and Sec. 5.4 claims 'immediate gradient instability' and 'NaN events' for baselines but provides no statistical aggregation (e.g., mean/std over seeds) or confidence intervals. Report results averaged over at least 3 random seeds to distinguish stochastic failure from systematic architectural flaws.
- **[science]** The benchmark evaluation (Sec. 5.2, Tab. 1) relies on a single run per model ('one generated video per benchmark scene'). For metrics like RotErr and TransErr, report the standard deviation across the 80 scenes and, if possible, multiple seeds to establish the statistical significance of the reported improvements over baselines.
- **[science]** The claim of '36x higher throughput' (Abstract) and efficiency gains in Tab. 1 lacks error bars or variance reporting. Provide the standard deviation of latency/throughput measurements across multiple inference runs to confirm the stability of these efficiency claims.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high level of technical sophistication, and the writing is generally clear and professional. However, there are specific areas where the text requires polishing to meet the standards of a final public preprint.

First, the LaTeX source contains numerous author-specific debug macros (e.g., `\cjs`, `\enze`, `\yy`, `\jc`, `\jy`, `\cai`, `\crh`, `\muyang`, `\haozhe`, `\haoyi`) and placeholder commands (`\tbd`, `\nan`, `\ph`, `\change`, `\bst`, `\snd`). These are clearly internal collaboration tools intended for the drafting phase. Their presence in the final manuscript is unprofessional and distracting. All such macros must be removed, and any text wrapped in them must be finalized or deleted before publication.

Second, in Section 3 (Method), the transition between the definition of token-wise GDN and the introduction of frame-wise GDN is abrupt. The subsection header "From Token-wise GDN to Frame-wise GDN" is followed immediately by the equation for token-wise GDN, then a paragraph explaining the limitations, and then the frame-wise equation. A brief transitional sentence is needed to explicitly state *why* the shift to frame-wise scanning is necessary for video modeling (e.g., to reduce the recurrent state update frequency from every token to every frame, thereby improving efficiency).

Third, in Section 5 (Experiments), under "Deployment efficiency path," the sentence "Note that the attention-sink variant means that we use the first latent frame as the attention sink with local window attention on softmax attention layers only, so that the memory consumption remains constant" is grammatically awkward and slightly confusing. It mixes the definition of the variant with its mechanism and result in a single, dense clause. Rephrasing this to clearly separate the mechanism (using the first frame as a sink) from the outcome (constant memory) would improve readability.

Finally, while the flow is generally good, some sentences in the Introduction are quite long and dense. For instance, the sentence beginning "Recent open-source systems achieve minute-scale..." could be split to improve readability. Overall, the paper is well-written but requires these specific edits to ensure clarity and professionalism.

Action items.

- [**writing**] Remove all author-specific debug macros (e.g., \cjs, \enze, \yy, \jc, \jy, \cai, \crh, \muyang, \haozhe, \haoyi) and the \tbd, \nan, \ph, \change, \bst, \snd commands from the final manuscript. These are clearly internal collaboration tools and must not appear in a public preprint.
- [**writing**] In Section 3 (Method), the phrase 'From Token-wise GDN to Frame-wise GDN' introduces a subsection but the text immediately following it jumps into equations without a clear transitional sentence explaining the motivation for the shift in granularity. Add a brief sentence bridging the token-level definition to the frame-level adaptation.
- [**writing**] In Section 5 (Experiments), the sentence 'Note that the attention-sink variant means that we use the first latent frame as the attention sink...' is grammatically clunky and slightly ambiguous. Rephrase for clarity, e.g., 'In the attention-sink variant, the first latent frame serves as the global attention sink, while local window attention is applied to softmax layers to maintain constant memory consumption.'